

Math 1314 Lesson 26

Use a calculator.

1) Find $\log_{10} 4 = \log 4 =$ _____

2) Find $\log_{10} 0.224 = \log 0.224 =$ _____

3) Find $\log_{10} 125 = \log 125 =$ _____

4) Find $\log_e 4 = \ln 4 =$ _____ (Note: $e = 2.718281828459$)

5) Find $\log_e 0.224 = \ln 0.224 =$ _____ (Note: $e = 2.718281828459$)

6) Find $\log_e 125 = \ln 125 =$ _____ (Note: $e = 2.718281828459$)

7) Find $\log_5 4 = \frac{\ln 4}{\ln 5} =$ _____

8) Find $\log_5 0.224 = \frac{\ln 0.224}{\ln 5} =$ _____

9) Find $\log_5 125 = \frac{\ln 125}{\ln 5} =$ _____

10) Find $\log_{12} 4 = \frac{\ln 4}{\ln 12} =$ _____

11) Find $\log_{12} 0.224 = \frac{\ln 0.224}{\ln 12} =$ _____

12) Find $\log_{12} 125 = \frac{\ln 125}{\ln 12} =$ _____

Property of Log: $\log_{10}(a \cdot b) = \log_{10}(a) + \log_{10}(b)$ $\ln(a \cdot b) = \ln(a) + \ln(b)$

13) a) Find $\log_{10}(3 \cdot 5) =$ _____

b) Find $\log_{10} 3 + \log_{10} 5 =$ _____

14) a) Find $\log_e(31 \cdot 25) =$ _____

b) Find $\log_e 31 + \log_e 25 =$ _____

15) a) Find $\log_{10}(7 \cdot 8) =$ _____

b) Find $\log_{10} 7 + \log_{10} 8 =$ _____

Property of Log: $\log_{10}\left(\frac{a}{b}\right) = \log_{10}(a) - \log_{10}(b)$

$$\ln\left(\frac{a}{b}\right) = \ln(a) - \ln(b)$$

16) a) Find $\log_{10}\left(\frac{3}{5}\right) =$ _____

b) Find $\log_{10} 3 - \log_{10} 5 =$ _____

17) a) Find $\log_e\left(\frac{13}{15}\right) =$ _____

b) Find $\log_e 13 - \log_e 15 =$ _____

18) a) Find $\log_{10}\left(\frac{7}{8}\right) =$ _____

b) Find $\log_{10} 7 - \log_{10} 8 =$ _____

Property of Log: $\log_{10} a^x = x \cdot \log_{10}(a)$

$$\ln a^x = x \cdot \ln(a)$$

19) a) Find $\log_{10}(4^3) =$ _____

b) Find $3 \cdot \log_{10} 4 =$ _____

20) a) Find $\log_e(6^3) =$ _____

b) Find $3 \cdot \log_e 6 =$ _____

21) a) Find $\log_{10}(8^5) =$ _____

b) Find $5 \cdot \log_{10} 8 =$ _____

22) a) Find $24 \cdot \ln \left[500 \left(1 + \frac{0.03}{12} \right) \right] = \underline{\hspace{2cm}}$ b) Find $\ln \left[500 \left(1 + \frac{0.03}{12} \right) \right]^{24} = \underline{\hspace{2cm}}$

23) a) Find $36 \cdot \ln \left[11000 \left(1 + \frac{0.06}{12} \right) \right] = \underline{\hspace{2cm}}$ b) Find $\ln \left[11000 \left(1 + \frac{0.06}{12} \right) \right]^{36} = \underline{\hspace{2cm}}$

24) a) Find $48 \cdot \ln \left[20000 \left(1 + \frac{0.07}{12} \right) \right] = \underline{\hspace{2cm}}$ b) Find $\ln \left[20000 \left(1 + \frac{0.07}{12} \right) \right]^{48} = \underline{\hspace{2cm}}$

25) Suppose you invested \$10,000 in a money market fund that earns 5% interest rate, compounded monthly. How long (years) will it take to accumulate \$15,000?

26) Suppose you invested \$25,000 in a money market fund that earns 4% interest rate, compounded monthly. How long (years) will it take to accumulate \$35,000?

